Statistical data analysis project (or Assignment 2)

This assignment was locked 27 Oct 2024 at 23:59.

Overview

The second assignment is simple. I want you to think of an interesting statistical question (s), find some open data, and use your knowledge gained during the course to answer your question (s). This is your opportunity to demonstrate all that you have learned during this course. You will be awarded (with marks) the clearer you demonstrate your skills. You must use Open Data because this assignment aims to turn your project into a slideshow presentation (using RMarkdown) [you can include it in an ePortfolio if you like].

Groups registration

Students are permitted to work **individually** or **in groups of up to 3** members for this Assignment.

Please note that the group sign-up for the Project is open for you all to choose a group.

Please click on the "People" tab on the left side of the canvas of this course and then click on the " Statistical data analysis project (or Assignment 2)" tab located at the top of that page. Then select a group and sign up for it.

This assignment is worth 25% and must be uploaded in PDF format in Canvas by **20.10.2024**.

Open Data

The data to be used must be open and ideally have a Creative Commons Licence. This will ensure you can share your work with anyone provided you make proper attribution. If you're not sure if the data is Open, contact the provider, and read the documentation. Some open data sources are provided below, but I encourage you to find others:

- https://www.kaggle.com
- UCI Machine Learning Repository
- data.gov
- world bank
- amazon web services
- google data sets
- youtube video data sets
- analytics vidhya
- quandl
- driven data
- http://www.abs.gov.au/

- https://www.data.vic.gov.au/
- http://www.bom.gov.au/

If you like you can also collect your data which takes significant time (You won't receive extra marks whether you gather your data or utilize the existing open data.). If you choose to collect your data, then explain how you collected your data and explain the sampling method. There should be enough detail here so that someone else could replicate your data collection.

Submission Instructions

The Assignment 2 report must be completed using the R Markdown template, feel free to use any template that you like.

Here is an example **MATH1324_Assignment_2_Template.Rmd**

Note that this is a slideshow template. The template includes basic instructions. You can read more **here**.

The slideshow presentation will be in a reproducible R Markdown format with written sections, R code, and output. **Presentations are limited (maximum) to 20 slides. Finally, for the submission, we only accept PDF format and we will not mark other formats. Online (or oral) presentations such as recording video and so on are not required for this assignment.** The presentation should compose of the following sections, and you can add more.

1- Presentation title and group/individual details [Plain text]: You can add the title of your presentation and student(s) details by updating the “title” and “author” entries at the top of the R Markdown Template.

2- Introduction [Plain text] or Problem Statement [Plain text]: State the overall problem/question driving the investigation. A good introduction provides a brief background to the problem, defines important terms, and leads to a strong rationale.

3- Data [Plain text]: If you collected your data, explain how you collected your data. There should be enough detail here so that someone else could replicate your data collection. Explain the sampling method if known. Ensure you reference the data source if you have used Open Data. List and explain the important variables. Explain everything that you do to preprocess the data.

4- Descriptive Statistics and Visualisation [Plain text & R code & Output]: Summarise the important variables in your investigation. Use visualisation to highlight interesting features of the data and tell the overall story. Explain how you dealt with data issues (if any), e.g. missing data and outliers.

Between Task 5, 6, and 7, you can choose any two of them for your assignment. You do not need to include all three tasks.

5- Hypothesis Testing and Confidence Interval [Plain text & R code & Output]: Apply an appropriate hypothesis test and confidence interval for your investigation. Ensure you state the hypotheses and check any assumptions. Report the appropriate values and interpret the results.

6- Categorical association [Plain text & R code & Output]: Apply an appropriate categorical association for your investigation. Report the appropriate values and interpret the results.

7- Regression analysis [Plain text & R code & Output]: Apply an appropriate Regression analysis for your investigation. Ensure you state the hypotheses and check any assumptions. Report the appropriate values and interpret the results.

8- Discussion [Plain text]: Discuss the major findings of your investigation. Discuss any strengths and limitations. Propose directions for future investigations. This is a good place to re-state your findings as a conclusion. What is the one take-home message the reader should leave with?

9- References [Plain text]: Provide a list of any references you use in the presentation.

Extension and special consideration

All the courses in your program follow the RMIT University Assessment policy for extensions and special consideration. Ensure you understand these guidelines (**https://www.rmit.edu.au/students/my-course/assessment-results/special-consideration-extensions/extensions**) before applying. (**https://www.rmit.edu.au/students/student-essentials/assessment-and-exams/assessment/extensions-of-time-for-submission-of-assessable-work**)

If you are prevented from submitting an assessment on time by circumstances outside your control, you may apply for an extension of up to seven (7) calendar days in writing (email) to your Course Coordinator (laleh.tafakori@rmit.edu.au) one working day BEFORE the due date [Working days are Monday to Friday].
Examples of the type of circumstances that may be considered include:

1. unexpected short-term physical or mental illness;
2. primary carer responsibility for a family member with an unexpected illness;
3. an unexpected, unavoidable employment commitment;
4. other unexpected personal circumstances outside your control, such as bereavement, being the victim of a crime or other trauma, severe disruption to living arrangements, or financial hardship, for instance, the sudden loss of employment or income.

Falsifying documents is considered fraud and treated very seriously. If you are proven to have falsified supporting documents for ANY REASON, you will likely have your course result converted to fail or other penalties applied. As a student, you must be aware of your obligations and responsibilities under RMIT Student Conduct (**https://www.rmit.edu.au/students/student-essentials/rights-and-responsibilities/student-responsibilities/conduct**). Your Course Coordinator will notify you (via email) that your extension has been approved and indicate the new due date if applicable.

If you require an extension of time longer than 7 days, you must apply for Special Consideration. An application for special consideration is made in advance of an assessment wherever possible, but normally within five working days after the assessment date.

To request an extension, please submit it through the "Assignment Extensions" form from the course navigation panel on the left.

Late Submission of Assessment

Assignments received late and without prior extension approval or special consideration will be penalized by a deduction of 10% of the total score possible per calendar day for that assessment.

For example: An assignment worth 40 marks will have 4 marks deducted per calendar day after the due date.

Collaboration vs. Collusion and Plagiarism/ Academic Integrity and AI tools:

Assignments will be submitted through Turnitin. **You need to ensure that your report's similarity is less than 30%, according to the Turnitin similarity report.** You are free to discuss the main aspects of the assignment with your classmates. However, keep in mind that this is an individual assignment, and you should demonstrate your effort and understanding. Because assignments will be submitted through Turnitin, all the material you submit will be checked for plagiarism. If plagiarism is detected, both the copier and the student copied from will be responsible. It is your responsibility to ensure you do not copy or do not allow another classmate to copy your work. If plagiarism is detected, both the copier and the student copied from will be responsible. It is good practice to never share assignment files with other students. You should ensure you understand your responsibilities by reading the RMIT University website on **academic integrity**.

- Go to the **Academic integrity page** for resources, study support, and contacts.
- Complete the **Academic Integrity Awareness credential**. This is the best way to understand what academic integrity is, how to maintain it, your responsibilities, and how to protect yourself from accidental breaches.
- **Know how to reference** including how to reference AI.

Kind regards,

Laleh

Criteria	Ratings					Pts
Introduction	**2 Pts** **Does everything almost correct** Background builds a strong and interesting rationale driving the investigation. Important concepts are detailed. The reader wants to know the answer	**1.33 Pts** **Has some mistakes but not too bad** Background and rationale driving the investigation were provided, but some minor details were missing/not clear.	**0.67 Pts** **Needs to review all the related contents carefully** A simple background and rationale to the investigation were provided, but it lacked detail/clarity.	**0 Pts** **Not acceptable** No background to investigation and/or rationale. The aim of the investigation was not clear.		2 pts
Data and descriptive statistics analysis	**4 Pts** **Does everything almost correct** The data description and descriptive statistics analysis was clear, concise and detailed. The presenters clearly had an in-depth knowledge of the data.	**2.67 Pts** **Has some mistakes but not too bad** The Data and descriptive statistics analysis was described in detail, but some minor aspects were not entirely clear/correct.	**1.33 Pts** **Needs to review all the related contents carefully** Important insight into the nature of the data and Data and descriptive statistics analysis was missing.	**0 Pts** **Not acceptable** The quality and nature of the data and Data and descriptive statistics analysis were unclear.		4 pts
methodology and Analysis	**11 Pts** **Does everything almost correct** The analysis is justified, assumptions checked, and results reported clearly. Results are presented and summarised meticulously. The interpretation of the results were clearly	**8.25 Pts** **Has some mistakes but not too bad** Most of the statistical tests and summaries were reported and interpreted correctly. There was some room for improvement in the analyses selected, reporting of	**5.5 Pts** **Needs to review all the related contents carefully** The analysis and presentation of results were mostly suitable, but some important elements (e.g. statistical tests, descriptive statistics or	**2.75 Pts** **Needs to study all the related contents from scratch** Major aspects of the analysis were inappropriate and/or completed incorrectly. Results of the statistical analysis are poorly presented and/or missing.	**0 Pts** **Not acceptable** Major aspects of the analysis were inappropriate and/or completed incorrectly. Results of the statistical analysis are poorly presented and/or missing. Results were misinterpreted.	11 pts

Criteria	Ratings					Pts
	and accurately communicated.	results and/or interpretations.	visualisations) were poorly described/ and or interpreted.	Results were misinterpreted.		
Discussion	**6 Pts** **Does everything almost correct** The results of the investigation are critically analysed and a conclusion was reached which related back to the initial investigation question. The discussion is succinct and conveys the take home message from the investigation.	**4.5 Pts** **Has some mistakes but not too bad** The results of the investigation are discussed in detail and an appropriate conclusion was reached. Minor details could be added, or some minor irrelevant points were included.	**3 Pts** **Needs to review all the related contents carefully** A discussion and conclusion are presented, however, there is a lack of insight into the results of the investigation. There is either insufficient discussion or the discussion raises many irrelevant points.	**0 Pts** **Not acceptable** There was no discussion or conclusion or the discussion and conclusion do not match the results. There is no attempt to interpret the results back to the context of the original investigation. The presenter did not appear to understand the implications of the investigation's findings.		6 pts
Presentation	**2 Pts** **Does everything almost correct** The presentation shows considerable effort and attention to detail. Its is succinct and difficult to fault.	**1.5 Pts** **Has some mistakes but not too bad** Overall, the presentation appears professional and well laid out. Only a few minor issues with crowding, appearance, readability, spelling/grammar or referencing.	**1 Pts** **Needs to review all the related contents carefully** The presentation has a number of issues with appearance, readability, spelling and/or referencing.	**0 Pts** **Not acceptable** The presentation appears messy, is difficult to read, has poor spelling/grammar, and poor referencing. The slide are crowded/empty.		2 pts

Total points: 25